NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Data Science Fundamentals [SC3021]

Chapter 1: What is Data Science?

Assoc. Prof. Melanie Herschel | CCDS

Course content

Data science overview

What is Data Science?

- **Definitions**
- **Misconceptions**
- **Differentiation from related fields**

Data Science Thinking vs. Computational Thinking

Data Science Ecosystems

Data preparation and data management

Data profiling

Structuring

Enriching

Data cleaning

Data management platforms

Schema design

Querying data

Data analytics and understanding

Descriptive analytics

Diagnostic Analytics

Prescriptive analytics

Predictive analytics

Data visualization

Design considerations

Privacy

Security

Ethics

Psychology

Policies

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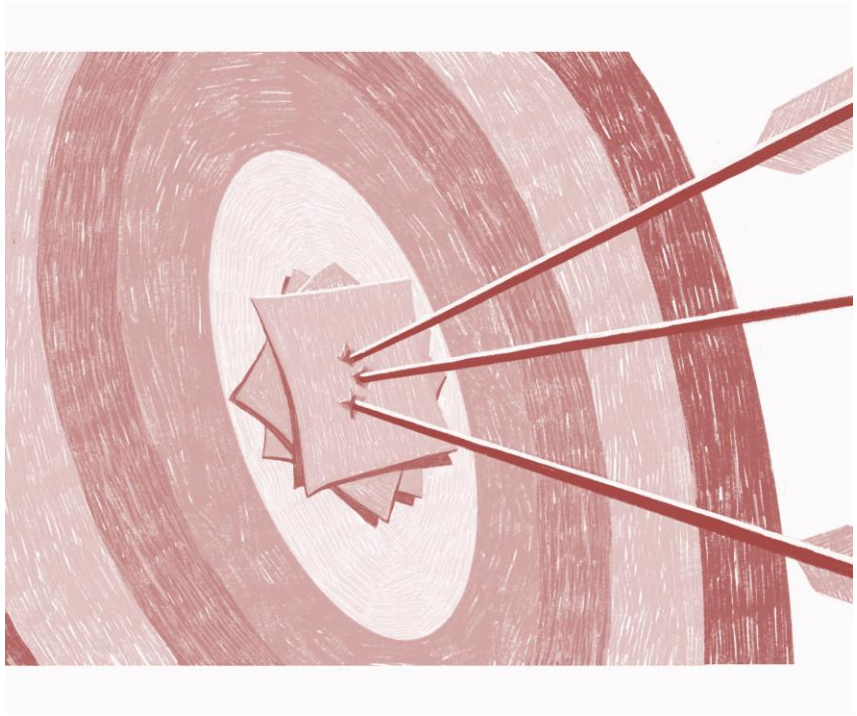
At the end of this lecture, you should be able to…

- …define and characterize data science.
- … explain how data science positions itself compared to computer science, maths, statistics, and AI.
- … argue when a problem is considered a data science problem or not.



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No one to rule them all…

DEFINITIONS OF DATA SCIENCE

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Take notes

Tuas Port to be world's largest fully automated terminal when completed in 2040

To boost efficiency, productivity and environmental sustainability, Tuas Port will have intelligent data-driven operations management systems and smart engineering and power management platforms.

StraitsTimes, 4/10/2019

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Take notes

Berkeley Public Health

News and Media / Research Highlights /

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How big data can save lives

Transforming medical care with machine learning

By Austin Price

6 min. read • Published August 25, 2019

"It's just algorithms layered on algorithms," says Professor Ziad Obermeyer. "Each of those algorithms parse out this super complex task into what is fundamentally a prediction."

machine learning—the process of running data through automated algorithms to produce a result or prediction—has become increasingly commonplace in our busy 21st century lives. Netflix uses algorithms to recommend movies based on what we've watched. Amazon uses algorithms to advertise items based on our purchase history. A self-driving car can approach a crosswalk and—after computing the world around it through light detection and ranging (LIDAR)—stop for a pedestrian.

"It's just algorithms layered on algorithms," says Professor Ziad Obermeyer. "Each of those algorithms parse out this super complex task into what is fundamentally a prediction."

A physician and researcher at Berkeley Public Health, Obermeyer conducts his research at the intersection of algorithms and the healthcare system. He joined the Berkeley faculty in 2018, after years of studying machine learning as a way to better

Berkeley Public Health, 25/18/2019

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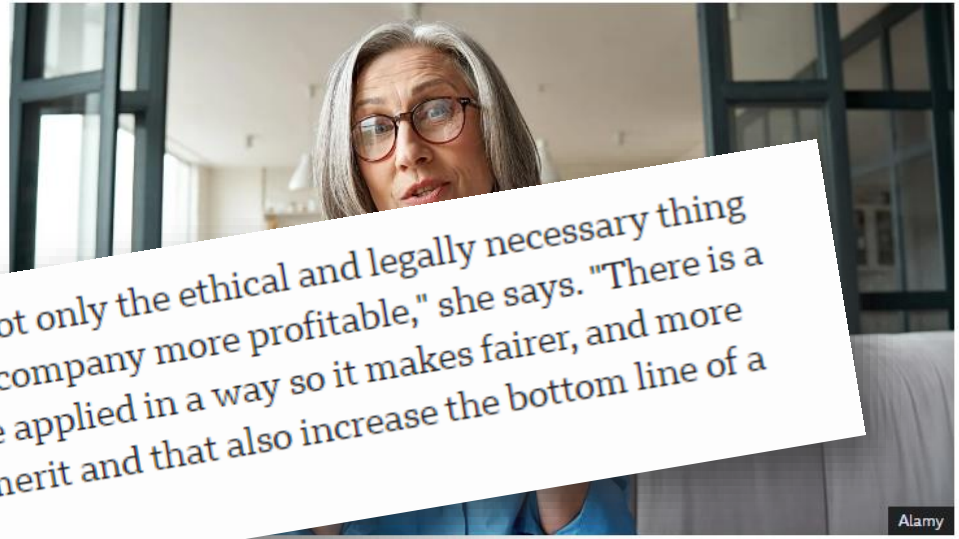
Take notes

AI hiring tools may be filtering out the best job applicants

16 February 2024

Share Save

Charlotte Lytton
Features correspondent



...nology, such as one-way video interviews, can leave some highly qualified candidates without the ...erviews they deserve (Credit: Alamy)

As firms increasingly rely on artificial intelligence-driven hiring platforms, many highly qualified candidates are finding themselves on the cutting room floor.

"Having AI that is unbiased and fair is not only the ethical and legally necessary thing to do, it is also something that makes a company more profitable," she says. "There is a very clear opportunity to allow AI to be applied in a way so it makes fairer, and more equitable decisions that are based on merit and that also increase the bottom line of a company."

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Takeaways from examples

- **Various terms** are seemingly used interchangeably for similar problems being solved based on data.
- Problems tackled are from **diverse domains**.
- A **specific problem,** question, or application drives the development.
- **Opportunities** and **risks** associated with using data to derive insights.

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A little bit of History…

1960 – ’80s

Origins in the statistics community

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A little bit of History…

Evolution into a mature field with computer science advancements in knowledge discovery, data mining

1990s

1960 – '80s

Origins in the statistics community

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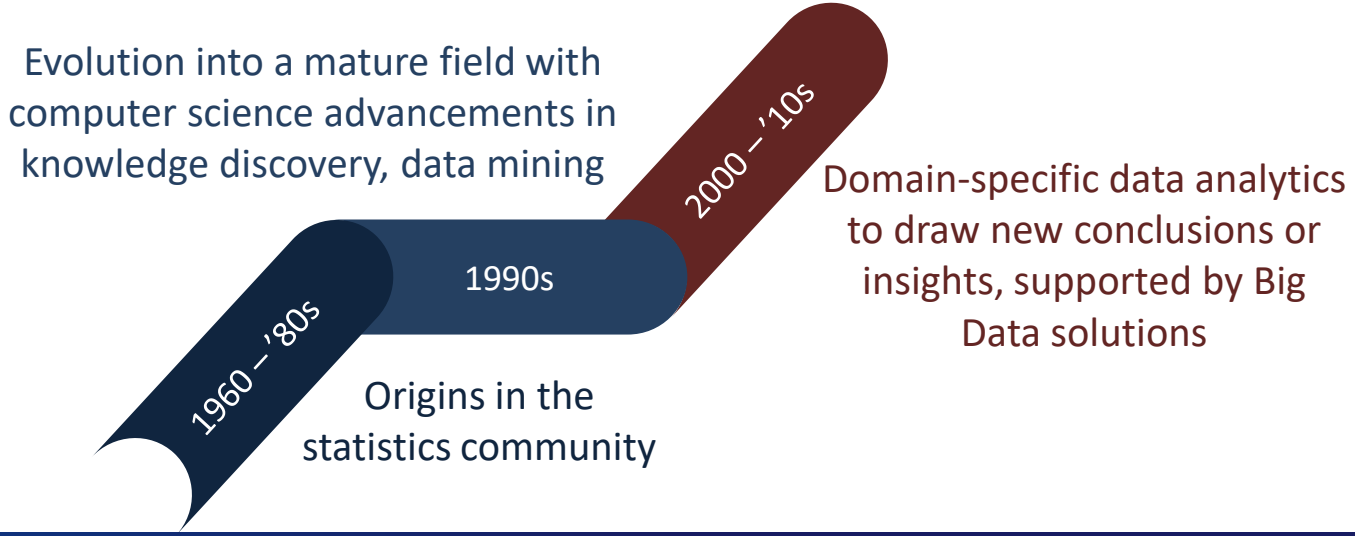
A little bit of History…



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A little bit of History…

Success of machine learning (deep learning)

Evolution into a mature field with computer science advancements in knowledge discovery, data mining

2010s

2000 – ’10s

Domain-specific data analytics to draw new conclusions or insights, supported by Big Data solutions

1990s

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Success of machine learning (deep learning)

2020s

Domain-specific, data-specific, and theoretical data analytics have formed the keystones of data science

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Definitions of Data Science

Process-based and goal-based perspectives

“ Data science encompasses a set of principles, problem definitions, algorithms, and processes for extracting non-obvious and useful patterns from large data sets. ” [MacKay00]

“ Systematic study of organization and use of digital data for research discoveries, decision-making, and data-driven economy. ” [Agrawal12]

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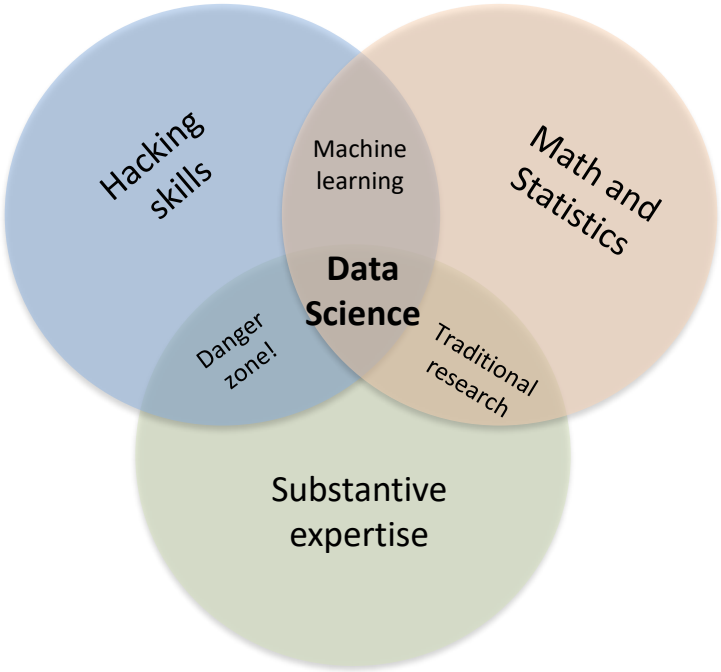
Definitions of Data Science

Disciplinary perspective

“ Data science comprises three distinct and overlapping areas: the skills of a statistician who knows how to model and summarize datasets […]; the skills of a computer scientist who can design and use algorithms to efficiently store, process, and visualize this data; and the domain expertise […] necessary both to formulate the right questions and to put their answers in context. ” [VanderPlas]



Adaptation of original figure by Drew Conway
http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagram

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Definitions of Data Science

Coverage of different relevant aspects

- Interdisciplinarity
- Data-based approach to problem solving
- Use of large and multi-modal data
- Focus on deriving insights and value by discovering patterns and relationships in data
- Underlying process-oriented life-cycle

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Working comprehensive definition

Used throughout this course

"Data science is a data-based approach to problem solving by analyzing and exploring possibly multi-modal and heterogeneous data, extracting knowledge and insight from it, and using information for better decision-making. It involves the process of collecting, preparing, managing, analyzing, explaining, and disseminating the data and analysis results." [Öszu23]

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Clarifications

What is data?

- A **representation of facts** or ideas in a formalized manner capable of being **communicated** or **manipulated** by some **process**.

Multi-modal and heterogeneous?

- **Multi-modal** data comprises different types of information (text, images, time series from sensors, tabular data, …)
- **Heterogeneous data** presents differences in syntax and semantics relating to represented facts (e.g., text vs XML vs CSV data or different attribute labels albeit having the same meaning).

Data-based vs data-driven?

- **Data-driven** is often understood as “we mainly (solely) rely on data to make a decision”. While data represents facts that data science may reveal, properly interpreting them is essential.
- **Data-based** indicates that data science supports decision making, without having its own agency.

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MISCONCEPTIONS

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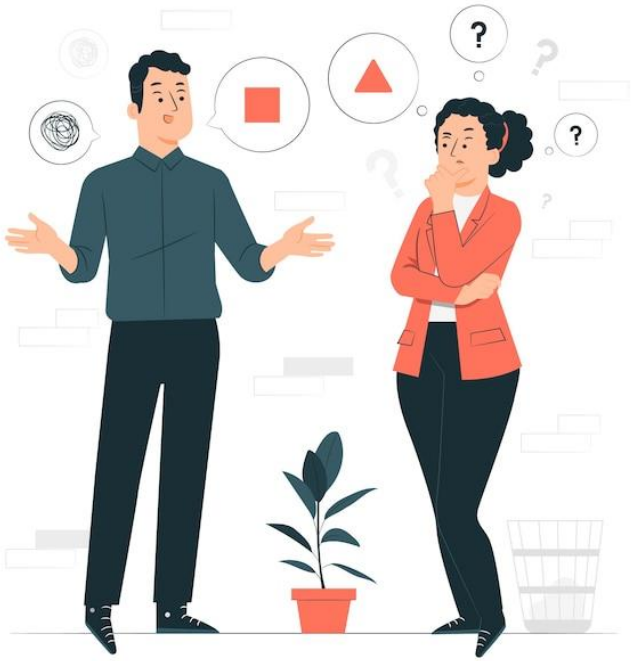
Misconceptions about data science

- Some possible reasons:
  - The diversity of definitions
  - The different disciplines involved
  - Overselling of the benefits
  - Strong focus on single aspects
- Knowing about common misconceptions is important
  - Supports understanding of what data science is and is not
  - Complements the general definition by stressing different relevant aspects



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✗ MYTHS

Data Science
=
Coding
(Python)

✓ FACTS

YES
coding skills are essential

BUT
data science is more than programming.

Beyond programming (and more generally Computer Science) skills, Data Science involves understanding statistics, data management, machine learning, and domain-specific knowledge to extract meaningful insights from data.

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✗ MYTHS

Data Science
=
Statistics

✓ FACTS

YES
a strong foundation in statistics is crucial

BUT
further skills are required by data scientists

Important Data Science aspects like collecting, preparing, managing, explaining, analyzing, and disseminating the data and analysis results are not (purely) solved by statistics.

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MYTHS

Data Science
=
Prediction

FACTS

YES
making predictions based on (trained) models is a significant component of data science
BUT
data science comprises a whole ecosystem to support this type of analytics.

More generally, data science encompasses further types of analytics and involves tasks such as data exploration, data cleaning, visualization, etc.

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✗ MYTHS

Data Science solves everything

✓ FACTS

YES
Data Science can provide valuable insights in many domains

BUT
there are no guarantees and expertise is required for correct interpretation.

Careful considerations on data adequacy, data quality, and domain-specific interpretation are some key factors!

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MYTHS

Data Science only applies on large datasets

FACTS

YES
large volumes of data are necessary for some methods

BUT
Data Science equally applies on small datasets

The focus of Data Science is on extracting value from the available data, small or big. Depending on the available data, different challenges need to be tackled as part of a Data Science solution.

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Summary of misconceptions

- Various reasons exists for misconceptions about what data science is.
- The definition we elaborated allows to debunk such misconceptions.
- It is part of a data scientist's work to clarify, rectify, and educate different stakeholders to manage their expectations or make them appreciate the broad work needed to devise a data science solution.

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Data science vs computer science, statistics, maths, AI

DIFFERENTIATION

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Differentiation overview

- Being multi-disciplinary, data science is related to multiple fields
- We focus on comparing data science to
  - Computer science
  - Mathematics
  - Statistics
  - Artificial intelligence(AI)

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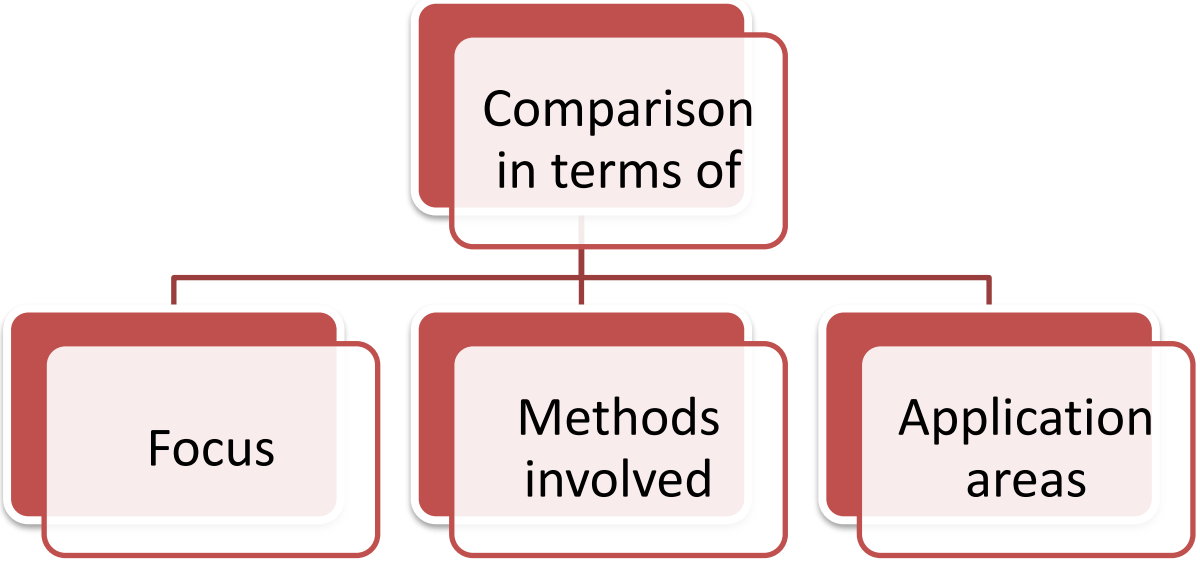
Differentiation overview



Lists are not exhaustive but provide prominent examples

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Data science

Focus: Application-specific extraction of insights and knowledge from data

Methods: data cleaning, data profiling, statistics, machine learning, visualization

Applications: Business intelligence, predictive analytics, scientific discovery, machine learning models, data-based decision-making

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Computer science

Focus: Design, development, and implementation of computer systems and software

Methods: algorithms, data structures, software engineering, programming languages, computer architecture

Applications: Software development, web development, cybersecurity, artificial intelligence, database management systems.

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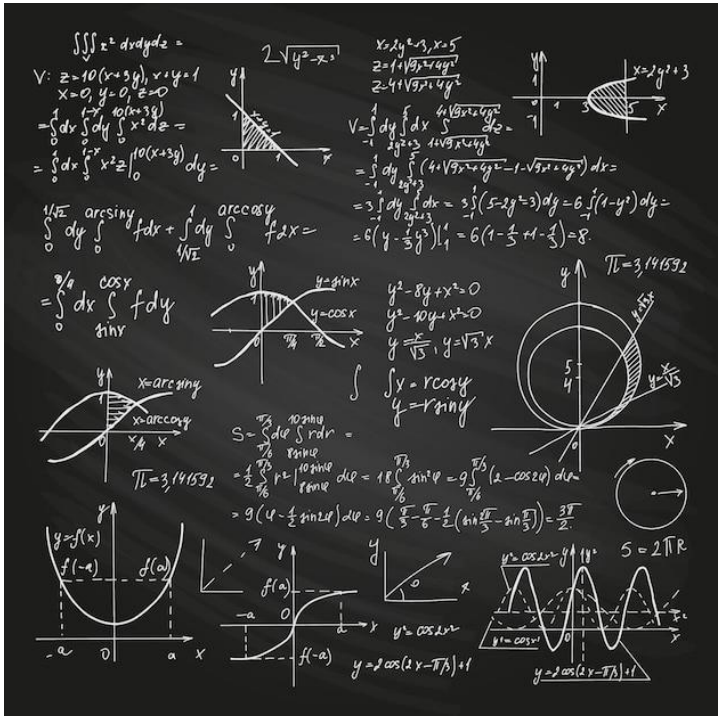
Mathematics



Focus: the study of numbers, shapes, and patterns

Methods: Algebra, geometry, calculus, discrete mathematics, trigonometry

Applications: engineering, physics, finance, economics, computer science

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Statistics

Focus: Collection, analysis, interpretation, presentation, and organization of data

Methods: statistical inference, hypothesis testing, regression analysis, experimental design

Applications: Research, quality control, market research, social sciences, medical research

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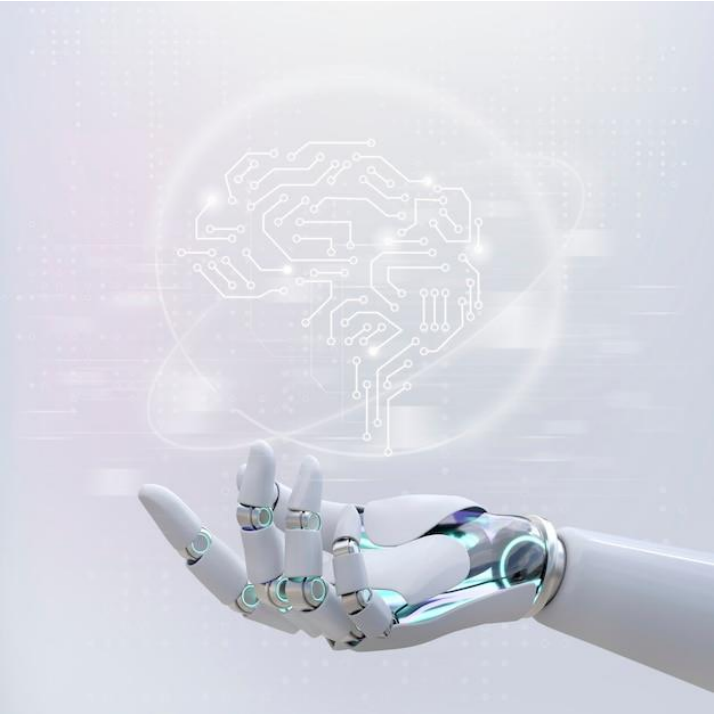
Artificial intelligence (AI)



Focus: Intelligent agents

Methods: machine learning, deep learning, natural language processing (NLP), computer vision, robotics

Applications: AI assistants, autonomous cars, medical diagnosis, game playing

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Comparison summary

	Data science	Computer science	Statistics	Mathematics	Artificial intelligence
Focus	Data insights	Computer systems	Data analysis	Mathematical concepts	Intelligent agents
Methods	Data cleaning, statistics, machine learning, visualization	Algorithms, programming, software engineering	Statistical inference, hypotheses testing, data organization	Algebra, geometry, calculus, trigonometry, Discrete math	Machine learning, natural language processing, computer vision, robotics
Applications	Business, research	Software development	Research, business	Various fields	Autonomous systems, AI assistants

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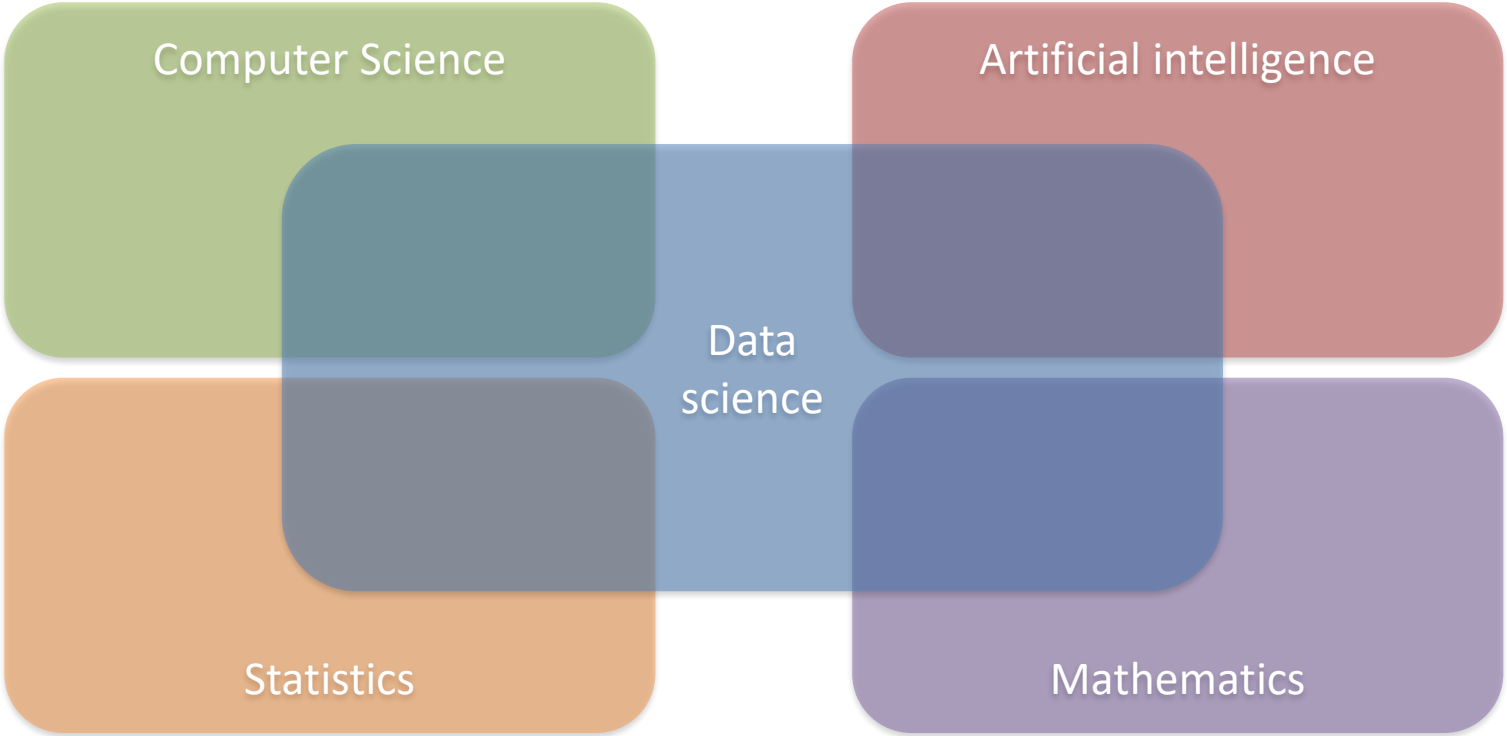
Data science-centric summary



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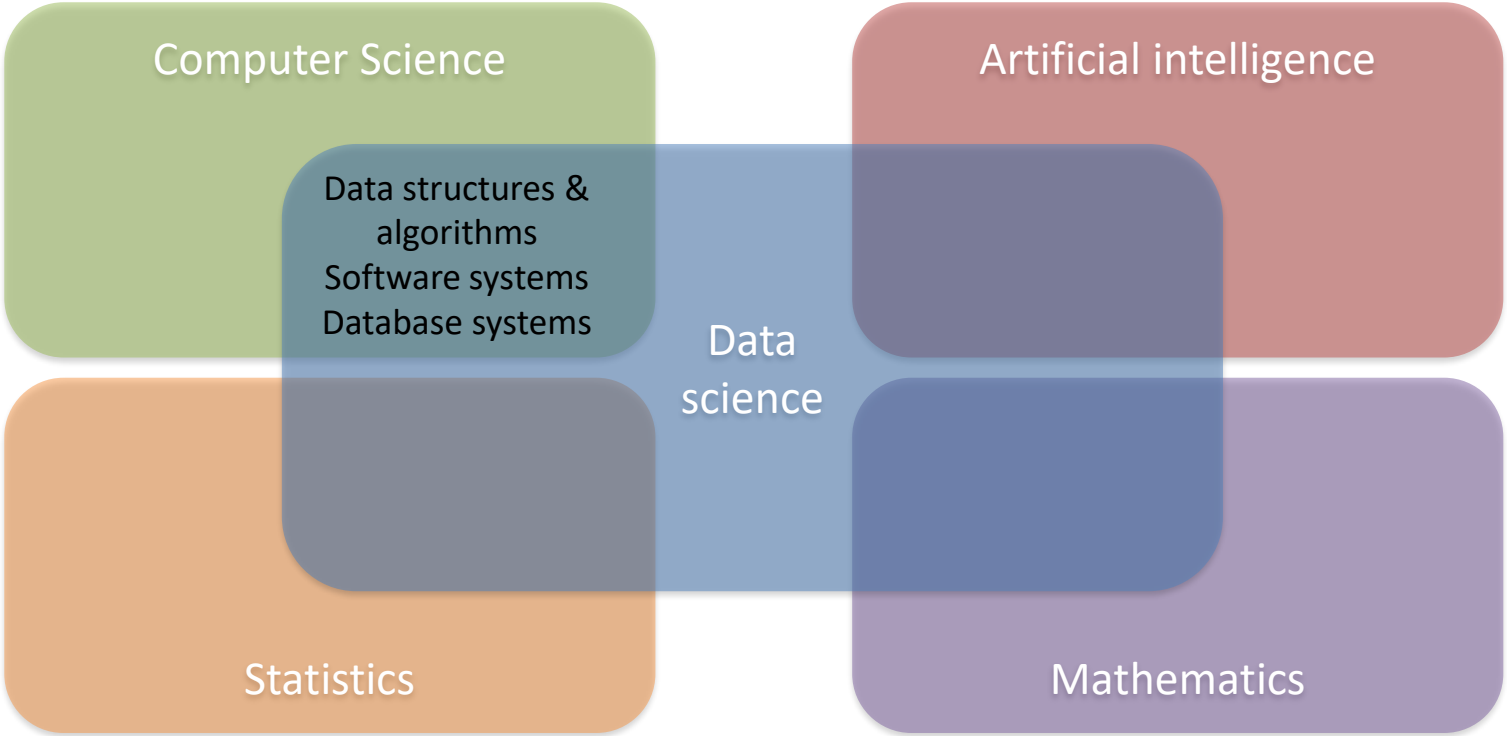
Data science-centric summary



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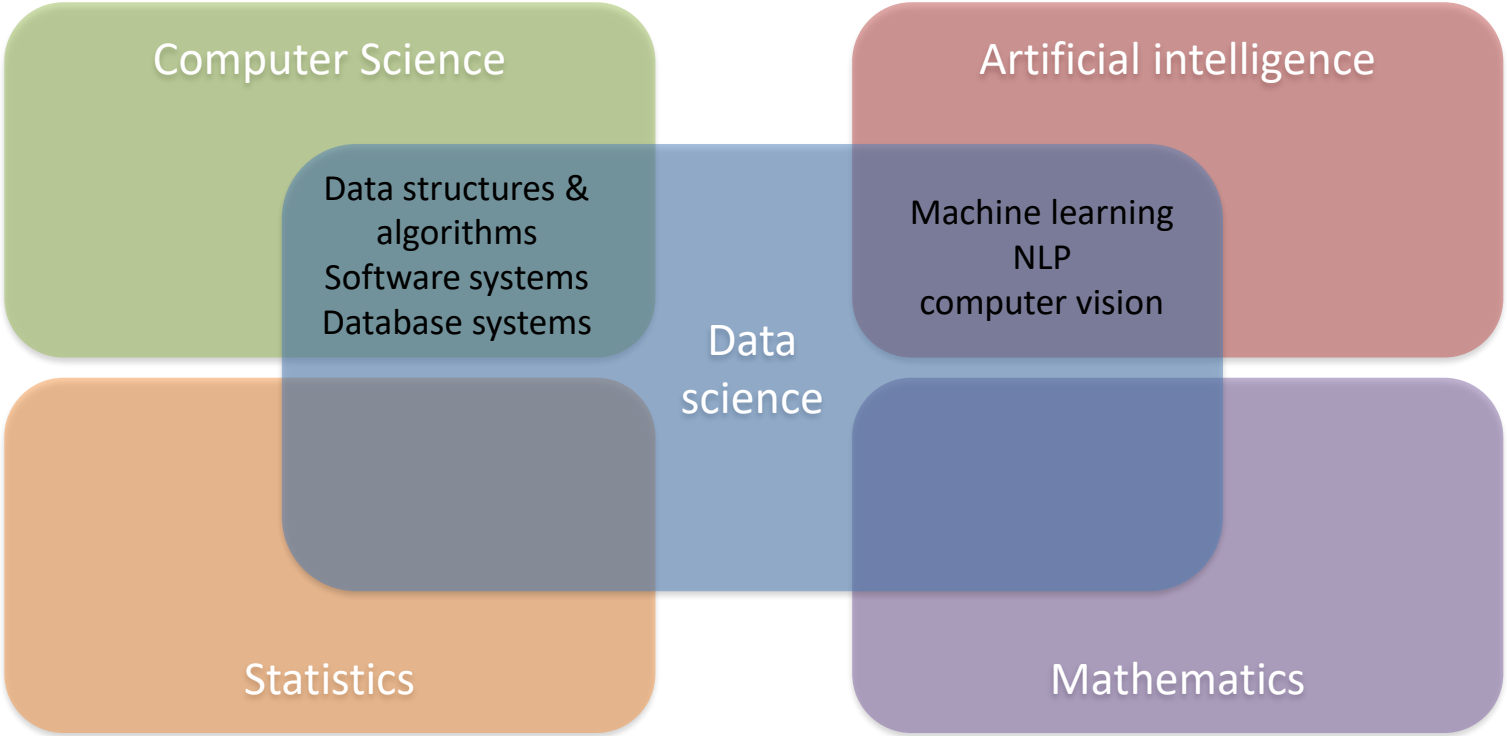
Data science-centric summary



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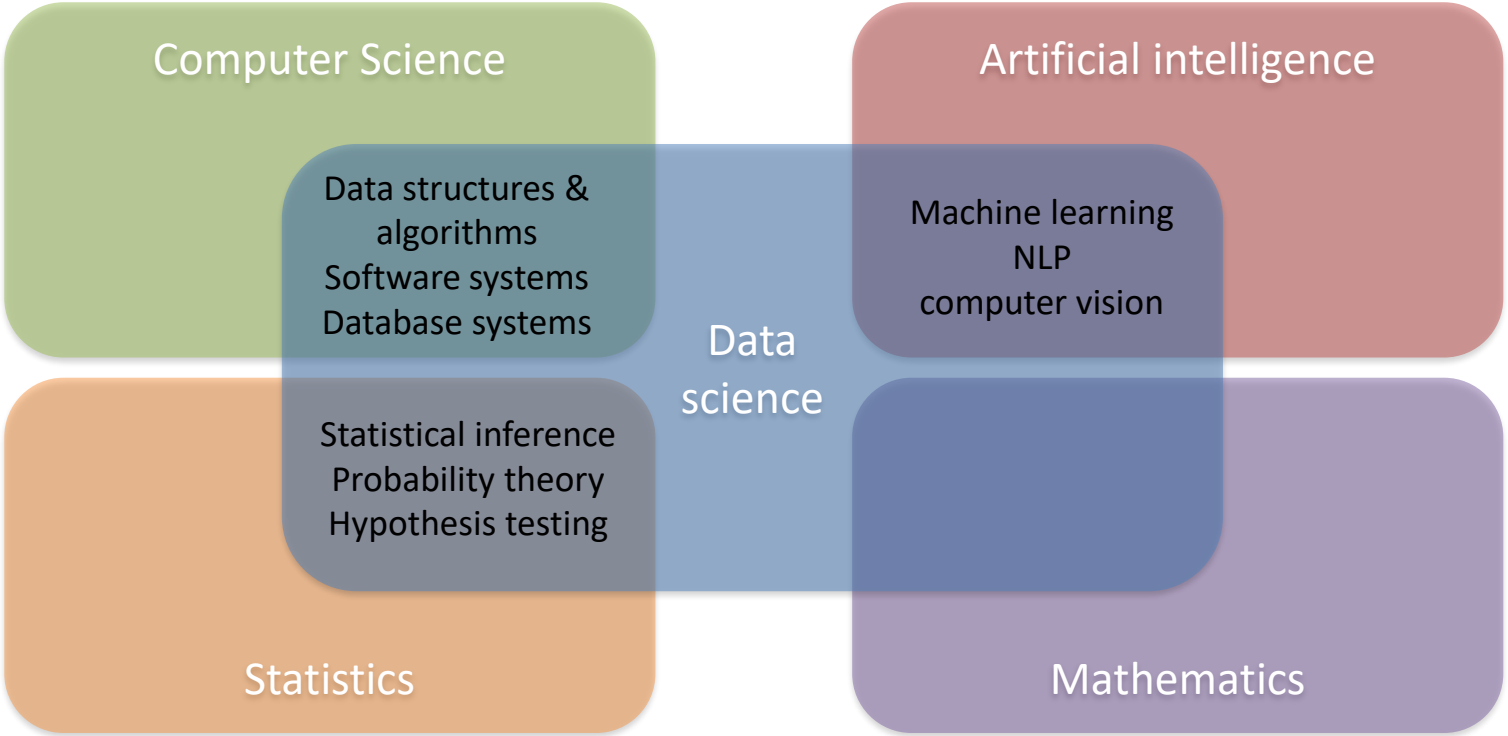
Data science-centric summary



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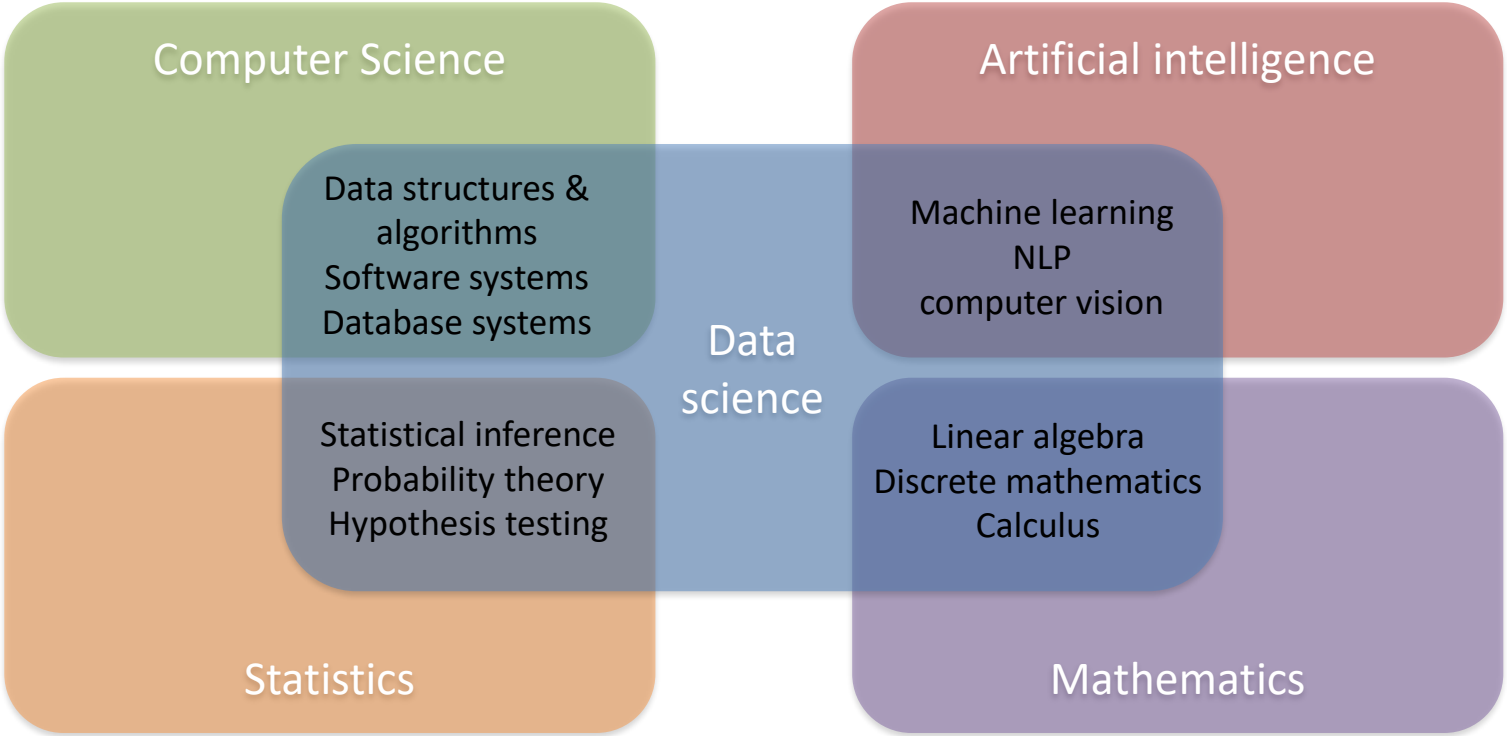

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SUMMARY

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Data science problems

- Are driven by a **domain-specific question** to be solved based on data
- Require **several stages** of processing the data (e.g., data collection, analysis, visualization)
- Rely on **methods from different but related disciplines** chosen based on the available data and problem to be solved.
- Are frequently ill-defined due to **misconceptions** about data science.
- **No one size fits all** solution, careful engineering throughout the whole process.
- This course will equip you with the necessary foundational knowledge and basic coding skills to **technically address data science problems end-to-end**.

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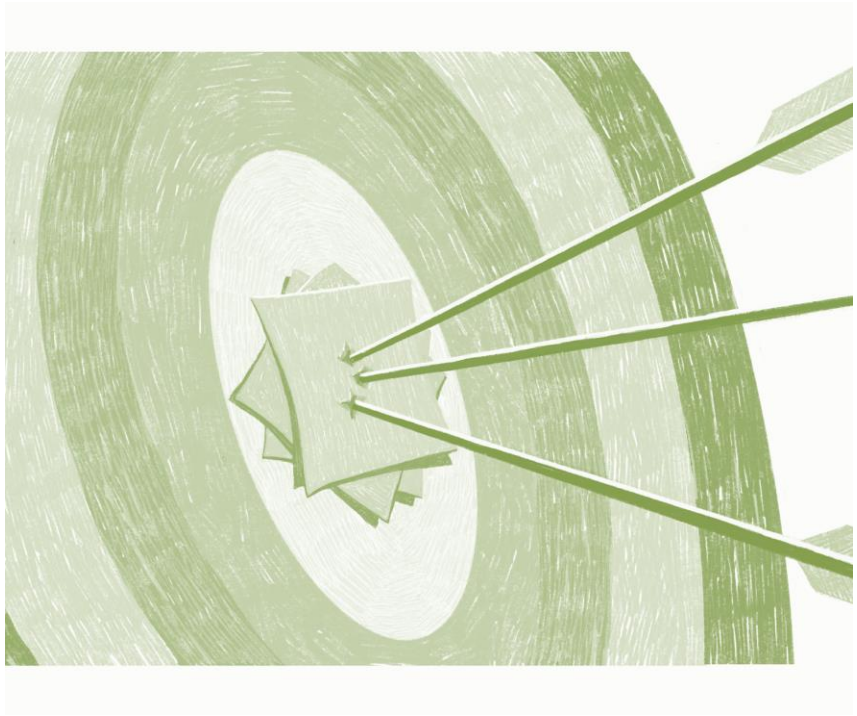
You should now be able to…

- …define and characterize data science.
- … explain how data science positions itself compared to computer science, maths, statistics, and AI.
- … argue when a problem is considered a data science problem or not.



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